QUESTIONS 5

1. Let $f(x) = xlnx$. Approximate $f(2)$ by the Hermite interpolating polynomial using $x_0 = 1$ and $x_1 = e$ and compare the error.($e \approx 2.7...$)

2. A natural cubic spline S on $[0, 2]$ is defined by

$$S(x) = \begin{cases} S_0(x) = 1 + 2x - x^3, & 0 \le x \le 1 \\ S_1(x) = 2 + b(x-1) + c(x-1)^2 + d(x-1)^3, & 1 \le x \le 2 \end{cases}$$

Find b, c, and d.